

Surface Chemistry as Boundary Geometry and Wake Interactions

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ABSTRACT

Surface chemistry describes adsorption, desorption, catalysis, and electron transfer as if these are properties of atoms sitting on a surface. They are not. Every one of these phenomena is an expression of boundary geometry — the shape, curvature, and periodicity of the termination where bulk periodicity ends — and the wake fields that geometry generates in the adjacent medium. When a crystal surface terminates bulk periodicity, it does not simply expose a layer of atoms. It imposes a geometric boundary condition on the electron density and vibrational field of the material. Those fields respond by generating structured disturbances — wakes — that propagate into the adjacent medium. The geometry of those wakes determines which molecules adsorb, where they adsorb, how tightly they bind, and what reaction pathways are available to them.

Adsorption is geometric locking: an incoming molecule's electron density geometry matches the surface wake geometry, and the molecule settles into the potential trough the wake defines. Desorption is geometric unlocking: thermal or photonic excitation oscillates the wake geometry until the lock breaks. Catalysis is wake-shaping: the catalyst boundary pre-distorts reactant geometry toward the transition-state geometry, reducing the energy cost of reaction. Electron transfer is wake-mediated routing: electron density flows along high-gradient channels in the wake field network rather than hopping across a gap. Poisoning, sintering, and surface reconstruction are all geometric failure modes — disruptions of the wake structure that degrade reactivity.

This paper argues that every existing quantitative model that successfully predicts surface reactivity trends — the d-band model, Brønsted-Evans-Polanyi relations, the Sabatier volcano plot, Marcus theory — works because it is a proxy for boundary geometry. The framework is not a replacement for those models. It is the physical basis that explains why they work and where they fail. The design implication is direct: fix the geometry, fix the chemistry. Composition, doping, and synthesis conditions are tools in service of that geometric goal.

1. Introduction

Surface chemistry textbooks are organized around outcomes. They describe binding energies, reaction rates, and selectivities as though these are fundamental quantities — properties of surface atoms that can be tabulated, compared, and interpolated. What the textbooks do not do is trace those outcomes back to the mechanism that produces them. The mechanism is geometric. Surface chemistry is boundary physics, and until the field treats it as such, the design of surface-active materials will remain an empirical exercise dressed in theoretical language.

A surface is a cutoff. It is the spatial coordinate at which the periodic potential of the bulk crystal terminates. That termination is not a subtle perturbation. It is a hard discontinuity in the symmetry of the material's environment. Every atom in the surface layer sits in a fundamentally different geometric situation than any atom in the bulk: it is missing half its neighbors, its coordination sphere is broken, and the forces that the periodic bulk imposed on it from all directions now arrive from only one side. The surface layer is not "like the bulk, but with some missing atoms." It is a different geometric environment, and the physics it produces is qualitatively different from bulk physics.

This geometric discontinuity is the origin of every surface phenomenon. Electron density at the surface is not the same as in the bulk because the boundary condition on the electron wave functions has changed. Vibrational modes at the surface are not the same as bulk phonons because the periodic restoring forces have been asymmetrically truncated. Chemical reactivity at the surface is not the same as in solution or in the gas phase because the geometry of the field environment surrounding a surface-bound species is determined by the wake field that the boundary generates — not by the intrinsic properties of the atoms in isolation.

This paper states its thesis plainly and defends it without qualification: geometry determines reactivity. Not composition, not electronic structure in the abstract, not

molecular orbital symmetry taken independently of structural context. The geometry of the boundary and the wake fields that geometry generates are the primary determinants of every surface chemical phenomenon. Electronic structure and molecular orbital arguments remain valid, but they are derivative — they describe the consequence of the geometry rather than its cause.

Two concepts carry the argument. The first is boundary geometry: the shape, curvature, periodicity, and termination structure of the surface. Boundary geometry includes the average crystallographic structure of a surface — whether it is a close-packed (111) face or an open (110) face — but also the local deviations from that average: steps, kinks, adatoms, vacancies, and reconstructed domains. All of these are geometric features. All of them modify the local boundary condition. All of them, as a consequence, modify the local reactivity.

The second concept is wake interactions. A wake is a structured disturbance in a medium produced by the presence of a boundary. In fluid dynamics, a wake behind a ship or a cylinder in flow is familiar. In surface physics, the same physics operates in the electronic and vibrational fields rather than in a fluid. When a surface boundary condition is imposed on the electron density, the electron gas does not simply stop at the surface. It oscillates — it generates Friedel oscillations that propagate into the bulk and into the vacuum, structured by the geometry of the boundary that created them. Analogously, the phonon field of the crystal generates surface phonon modes — vibrational wake modes that exist because the boundary condition has broken the periodicity that defined bulk phonon dispersion. These wakes are not secondary effects. They are the medium through which the surface communicates its geometry to approaching molecules, to adsorbed species, and to the electron density of the near-surface region.

The structure of this paper follows the mechanism. Section 2 establishes surfaces as geometric cutoffs and explains reconstruction, relaxation, steps, and defects as geometric responses to the broken boundary condition. Section 3 develops the concept of surface wakes — what they are, how they form, and how their geometry is determined by boundary

shape. Sections 4 and 5 apply the framework to adsorption and desorption, respectively. Section 6 treats catalytic surfaces as wake-shaping boundaries. Section 7 argues that boundary curvature is the master variable for surface reactivity and that the d-band model is a geometric proxy. Section 8 addresses electron transfer as wake-mediated routing. Section 9 analyzes poisoning, sintering, and surface reconstruction as geometric failure modes. Section 10 describes diagnostic and engineering strategies for restoring geometric performance. The Discussion places this framework in relation to established models. The Conclusion states the design principle without qualification.

2. Surfaces as Geometry Cutoffs

2.1 The Bulk-to-Surface Transition as a Geometric Event

A crystal surface is the spatial coordinate where bulk periodicity terminates. That termination is a geometric event with specific physical consequences. In the bulk, every atom sits at a lattice site defined by the three-dimensional translational symmetry of the crystal. The potential energy surface it experiences is periodic in all three directions. The electronic wave functions are Bloch states — they extend through the entire crystal with a well-defined k -vector and energy. Phonons are propagating waves with a dispersion relation set by the bulk force constants and mass distribution. None of this survives the surface cutoff intact.

At the surface, translational symmetry is broken in the direction perpendicular to the surface. The Bloch condition no longer applies in that direction. Electronic states at the surface are no longer required to be extended in the perpendicular direction — they can be localized at the surface, decaying exponentially into both the bulk and the vacuum. These are Shockley surface states and Tamm surface states, and their existence is a direct consequence of the geometric boundary condition, not of any special chemistry of the surface atoms.^[1] The surface atoms have not changed their intrinsic electronic character;

the geometry of their environment has changed, and that geometric change is what produces localized surface electronic states.

Electron density at the surface is redistributed relative to the bulk. The charge density does not simply stop at the outermost atomic layer — it spills out into the vacuum, forming a dipole layer at the surface. The magnitude and spatial extent of this spillout is a function of boundary geometry. A close-packed surface with high atomic density has a different spillout geometry than an open surface with low atomic density. That spillout geometry is the outermost expression of the surface wake field, and it is what an approaching molecule encounters first.

Surface reconstruction and relaxation are the material's geometric response to the broken symmetry imposed by the boundary cutoff. Relaxation refers to the change in interlayer spacing at the surface relative to bulk values — the top layer typically contracts toward the second layer because it has lost the outward restoring force of the missing layer above.^[2] This is the lattice seeking a new geometric equilibrium given the new boundary condition. Reconstruction goes further: the surface unit cell changes its periodicity, sometimes dramatically. The Si(111)-(7×7) reconstruction, the Au(111)-herringbone reconstruction, and the Ir(100)-(5×1) reconstruction are all cases where the surface boundary condition imposes a geometric reorganization that the bulk would never undergo. These are not chemical reactions. They are geometric phase transitions driven by the surface boundary condition.

The physical point is this: reconstruction and relaxation happen because the bulk geometry is not the optimal geometry given the new boundary condition at the surface. The atoms rearrange to find a lower-energy configuration in the new geometric context. The fact that the stable surface geometry is often very different from a simple bulk truncation is evidence that the surface boundary condition is a strong constraint — strong enough to change the geometry of multiple atomic layers below the surface.^[3]

2.2 Curvature, Steps, and Defects as Geometric Amplifiers

A perfectly flat, defect-free surface is the simplest geometric case. Its boundary condition is uniform — the same geometry at every surface unit cell. Its wake field is uniform — Friedel oscillations with the same wavelength and amplitude everywhere. Its reactivity is accordingly uniform and, in practice, low. Flat surfaces are poor catalysts. This is not coincidental.

Real surfaces deviate from the flat ideal. They have steps — edges where one terrace ends and another begins at a lower level. They have kinks — irregularities along step edges where the step itself bends. They have adatoms — individual atoms sitting above the surface plane. They have vacancies — missing atoms in the surface layer. Each of these features is a local deviation in boundary curvature. Each one imposes a locally different boundary condition. Each one generates a locally different wake field. And each one is, invariably, more reactive than the flat terrace around it.

Step edges are regions of high local curvature. An atom at a step edge is coordinated differently than a terrace atom in two directions simultaneously — it is at the junction of two differently oriented surface planes. The boundary condition at a step edge is a geometric compound of two terrace geometries, and the wake field it generates has higher spatial gradient than either terrace alone.^[4] High gradient means steep potential walls around the wake trough, which means stronger geometric locking of incoming molecules. This is why step edges are preferential adsorption sites. It is not that step-edge atoms are chemically special — their electronic character is set by their coordination geometry, which is geometrically special.

Kink sites are higher curvature still. A kink atom at a step edge on a (100) surface has a coordination number of 7 in an FCC metal, compared to 9 for a terrace atom and 8 for a straight step-edge atom. Terrace atoms have the highest coordination, kink atoms the lowest. Coordination number is a proxy for geometric boundary condition: low coordination means the boundary has cut away more of the atom's environment, imposing a stronger geometric discontinuity and generating a stronger local wake gradient. The

Stranski-Krastanov layer growth model implicitly acknowledges this — kink sites are where adatom incorporation is kinetically preferred because the geometric locking at kinks is tighter than at terraces.^[5]

Vacancies work differently. A surface vacancy removes a node from the periodic boundary condition. The surrounding atoms relax inward, and the boundary geometry around the vacancy is distorted — there is a local depression in the surface potential. The wake field in the vicinity of the vacancy must route around the geometric hole the vacancy creates. This routing creates field gradient concentrations at the vacancy perimeter that can be highly reactive — particularly for molecules that are geometrically compatible with the vacancy's shape. Some catalytic reactions are specifically promoted by vacancy sites for exactly this reason: the vacancy geometry matches the transition-state geometry of the reaction more closely than the terrace does.

The general principle from steps, kinks, and vacancies is this: curvature amplifies the wake field gradient. Higher curvature means steeper field gradients. Steeper field gradients mean stronger geometric locking and higher reactivity. This is why surface scientists measure step density and why metal-on-metal growth at stepped substrates proceeds differently than on flat terraces. Curvature is not a secondary structural detail — it is the primary geometric variable that controls reactivity.

3. Wake Formation at Boundaries

3.1 What a Wake Is and Why Boundaries Generate Them

A wake is a structured disturbance in a medium produced by the presence of a boundary or a moving object. In fluid mechanics this is textbook material: a cylinder in a flow generates a wake downstream because the boundary condition at the cylinder surface imposes a constraint on the fluid velocity field, and the fluid responds by forming a structured pattern of recirculation and oscillation that persists well downstream of the

cylinder. The wake is not a mysterious phenomenon — it is the mandatory response of a continuous medium to a geometric boundary condition imposed on it.

At a solid surface, the same physics operates in two coupled fields: the electron density field and the phonon field. The surface imposes a boundary condition on both. The electron gas cannot penetrate into the vacuum without energy cost, so its density drops from bulk value to near zero over a distance of a few angstroms at the surface. But a uniform, abrupt drop is not a solution to the quantum mechanical boundary value problem — the electron density must oscillate before it decays. These oscillations are Friedel oscillations: density waves in the electron gas that are generated because the geometric boundary condition at the surface forces the wave functions into configurations that produce real-space charge density oscillations.^[6]

Friedel oscillations are surface wakes in the electron density field. They have a wavelength of approximately π/k_F , where k_F is the Fermi wave vector of the metal — a quantity that is set by the electron density, which is set by the crystal structure, which is set by the boundary geometry. The Friedel oscillations decay in amplitude with distance from the surface, but they do not decay instantly. They persist for several atomic layers into the bulk, and they extend a comparable distance into the vacuum side where they mediate the interaction between the surface and approaching adsorbates. An adsorbate approaching the surface does not encounter a uniform potential field — it encounters the spatially oscillating Friedel wake, and where it lands on that wake (at a trough or a crest, at a node or an antinode) is determined by the geometric match between the molecule's own electron density structure and the wake's spatial structure.

Surface phonons are the wake analog in the vibrational field. Bulk phonons are propagating vibrations whose dispersion relation — the relationship between frequency and wave vector — is set by the periodic bulk force constant matrix. At the surface, the periodicity is broken. The phonon boundary condition changes. New modes appear that have no bulk analog: Rayleigh waves (surface acoustic waves), optical surface phonons in polar crystals, and dipole-active surface phonon modes.^[7] These surface phonon modes are the vibrational

wake — vibrations that exist because the boundary condition at the surface has broken the symmetry that defined bulk phonon propagation. They carry energy laterally along the surface and couple the translational motion of the lattice to the vibrational state of adsorbed molecules.

The wake framework resolves a conceptual confusion that plagues surface chemistry: the idea that surface atoms are chemically special because they are "unsaturated" or have "dangling bonds." Dangling bonds are a geometric artifact — they are the electronic consequence of the geometric fact that the surface atom is missing coordination partners on one side. The reactivity of a dangling bond is not an intrinsic property of the atom that owns it; it is a property of the wake geometry at that site. Change the geometry — by adsorbing a passivating species, by reconstructing the surface, by applying strain — and the dangling bond's reactivity changes accordingly, even though the atom's intrinsic electronic character has not changed.

3.2 Wake Geometry as the Driver of Surface Reactivity

The wake field at a surface is not uniform unless the surface boundary geometry is uniform — which, as Section 2.2 established, it never perfectly is. The spatial structure of the wake is a direct map of the boundary geometry. This mapping is the central operative mechanism of surface chemistry, and understanding it quantitatively is what transforms surface science from an empirical correlation science into a predictive one.

A flat, periodic surface generates a wake with the periodicity of the surface lattice in the lateral directions and monotonic Friedel oscillation decay in the perpendicular direction. This is the simplest case. Every adsorption site on such a surface is geometrically equivalent, and the wake geometry at every site is the same. In practice this is approximated by large, well-ordered terraces on low-Miller-index single crystal surfaces — and these surfaces are known to be relatively low in reactivity compared to stepped or defective equivalents.

A stepped surface breaks the lateral uniformity of the wake. The step edges impose additional boundary conditions on top of the terrace periodicity, and the resulting wake field is a superposition of the terrace wake and the step-edge wake. The step-edge wake has shorter spatial wavelengths — higher Fourier components — because the step geometry is a sharper feature than the terrace corrugation. This means the step-edge wake has steeper gradients, and the potential troughs in the wake field are deeper at step edges than on terraces.^[4] The spatial structure of the wake directly explains the preferential adsorption at step edges that is universally observed in surface science experiments.

A curved nanoparticle surface has a wake with angular structure that depends on the particle's radius and faceting. For a 2 nm gold nanoparticle, the average coordination number of surface atoms is substantially lower than for a bulk surface, and the Friedel oscillations are not the plane waves of a flat surface — they are spherical-wave-like disturbances with angular nodes set by the particle geometry.^[8] The wake geometry of a small nanoparticle is fundamentally different from that of the corresponding bulk surface, and this difference accounts for the dramatically enhanced reactivity of small gold nanoparticles relative to bulk gold — a material that is chemically inert as a macroscopic solid but highly active as a nanoscale catalyst.^[9]

The key insight is that reactivity does not track composition — it tracks wake geometry. Two surfaces with identical elemental composition but different geometric structures have different wake fields and therefore different reactivities. Platinum (111) and platinum (557) have the same elemental composition but different geometric structures — (557) has a step density of one step per 12 surface atoms — and they have dramatically different reactivities for CO oxidation and N₂ dissociation. The difference is geometric, mediated by the wake field.

4. Adsorption as Geometry Locking

4.1 The Geometry Matching Mechanism

Adsorption is the process by which an atom or molecule from the gas phase or solution comes to rest on a surface in a configuration that is more stable than the free state. The textbook description of adsorption invokes orbital overlap — the molecular orbitals of the adsorbate interact with the electronic states of the surface, leading to bond formation. This description is not wrong, but it locates the mechanism at the wrong level. Orbital overlap between molecule and surface is the consequence of geometry matching — it occurs when and because the molecule's electron density geometry is compatible with the surface wake geometry at a particular site.

The approach of a molecule toward a surface is not a simple two-body interaction. The molecule enters the surface wake field at a distance of several angstroms — well before it reaches the first atomic layer. The wake field the molecule encounters is spatially structured, as described in Section 3. The molecule's own electron density has a specific geometric shape: it may be linear, spherically symmetric, tetrahedral, or something more complex. As the molecule descends through the wake field, it experiences a spatially varying potential that pushes and pulls on its electron density. The molecule will preferentially settle into a position and orientation where its electron density geometry is geometrically compatible with the wake field geometry — where its shape fits the wake's landscape.

This is what "adsorption site selectivity" means in physical terms. A CO molecule has a linear electron density geometry with high density at the carbon end and lower density at the oxygen end. It prefers atop adsorption sites on most transition metals because the wake field geometry at an atop site has cylindrical symmetry that matches the CO molecular axis and presents the steepest field gradient in the vertical direction, allowing CO to lock in with its carbon end pointing down toward the surface.^[10] Changing the surface geometry — by introducing steps, by alloying, by strain — changes the wake field geometry at the surface and can redirect CO to bridge or hollow sites, changing its binding energy and its subsequent chemistry.

Chemisorption and physisorption are not categorically different phenomena — they are different depths of geometric locking. Physisorption is shallow locking: the molecule's electron density geometry is only weakly perturbed by the surface wake. The molecule settles into the outer oscillations of the Friedel wake, where the potential is shallow and the gradients are weak. The molecule retains most of its gas-phase geometry and can desorb readily because the lock is shallow. Binding energies in physisorption are typically 10–50 meV — the depth of the outer wake troughs.^[u]

Chemisorption is deep locking: the molecule descends into the inner wake field, where gradients are steep and the potential trough is deep. In this regime the molecule's own electron density is substantially deformed — it has to change its geometry to conform to the wake field geometry. This deformation is what surface scientists call "charge transfer" and "rehybridization." An sp-hybridized CO molecule adsorbed in a chemisorbed state shows a modified C-O stretching frequency, a change in molecular orbital occupancy as detected by ultraviolet photoelectron spectroscopy, and a binding energy of 0.5–3.0 eV depending on the surface. These changes are all expressions of geometric deformation: the molecule has been forced into a geometry that differs from its gas-phase equilibrium because the deep wake trough it has locked into has a geometry that requires that deformation.

The binding energy of chemisorption is the energy cost of that geometric deformation minus the energy gained by the geometric locking. When the wake field geometry at a site is a good match for the molecule's geometry, the geometric deformation is small and the binding energy is dominated by the locking energy — a strongly bound but undistorted adsorbate. When the match is poor, the deformation cost is high and the binding energy is lower. This is the physical content of the frontier molecular orbital argument for adsorption: HOMO-LUMO matching is a symmetry statement about geometric compatibility between molecular and surface electronic geometry.

4.2 Site Selectivity from Geometric Matching

Adsorption site selectivity is a direct consequence of geometric wake matching. Different surface sites have different wake field geometries, and different molecules have different electron density geometries. The observed pattern of site preferences is not a database of empirical binding affinities — it is a mapping of geometric compatibilities.

Step edges and kink sites are preferentially occupied at low coverages on most surfaces because, as established in Section 2.2, their boundary curvature is higher and their wake field gradients are steeper. The potential troughs in the wake at these sites are deeper than on flat terraces. Any molecule with sufficient thermal energy to diffuse on the terrace will sample the terrace wake field until it finds the deepest available trough — which is at the step edge or kink site. Energetically, the molecule is executing a geometric optimization: it is finding the position on the surface where its geometry best fits the wake.

Different molecules prefer different sites on the same surface because their electron density geometries are different. Nitrogen (N_2) with a symmetric linear geometry and weak polarizability prefers high-symmetry hollow sites on $\text{Fe}(111)$ and $\text{Ru}(0001)$ — sites where the wake field has the 6-fold or 3-fold symmetry that matches the approach geometry for end-on N_2 adsorption.^[12] Benzene with a flat, aromatic electron density distribution prefers flat-lying adsorption on the close-packed (111) faces of transition metals — sites where the wake field has the lateral periodicity that meshes with benzene's 6-fold electron density geometry. The match is geometric.

The Sabatier principle — that catalytic activity is optimal when binding is neither too weak nor too strong — is a geometric statement in disguise. Too weak a binding means shallow locking: the wake field gradient at the surface is too low to hold the molecule long enough for reaction. Too strong a binding means the molecule locks so deeply into the wake field that it cannot deform into the transition state geometry — it is trapped in a local geometric minimum that is not the reaction path. The optimum is a geometric balance: the wake field depth is sufficient to hold the molecule but shallow enough to allow the geometric deformation required by the reaction pathway. The volcano plot, in the geometric framework, is a plot of wake field depth versus reaction rate, and the peak of the volcano

identifies the wake geometry that is optimally matched to both the reactant geometry and the transition state geometry of the specific reaction.^[13]

5. Desorption as Wake-Driven Release

5.1 The Unlocking Mechanism

Desorption returns an adsorbed species to the gas phase or solution. The thermodynamic description — the adsorbate must acquire enough energy to surmount the desorption barrier, which equals the binding energy for a non-activated adsorption process — is correct but mechanistically empty. It says what the energy requirement is; it does not say how the energy is delivered to the locked geometric configuration or how the lock actually breaks.

The mechanism is this. A molecule locked in the surface wake field sits in a potential trough defined by the wake geometry. The trough has a specific shape: steep walls set by the wake gradient, a minimum at the locking geometry, and a well depth equal to the binding energy. The locked molecule vibrates in this trough — it oscillates around the equilibrium locking geometry with an amplitude set by its thermal energy. At low temperatures, the amplitude is small and the molecule remains locked. As temperature rises, the vibrational amplitude increases.

Simultaneously, thermal excitation drives oscillations in the surface wake field itself. Surface phonons — the vibrational wake modes established in Section 3.1 — are thermally populated according to the Bose-Einstein distribution. As temperature increases, higher-frequency surface phonon modes become populated, and the amplitude of wake field oscillations at the surface increases. The wake trough is not static: it is oscillating at the surface phonon frequencies, its depth and shape fluctuating with the lattice vibrations.

Desorption occurs when the combination of adsorbate vibrational amplitude and wake trough oscillation amplitude is sufficient to cause the locked geometry to exit the trough.

Specifically, when the trough oscillation distorts the trough geometry enough that the adsorbate's locked geometry no longer fits — when the wake trough shape becomes incompatible with the adsorbate geometry — the lock breaks and the molecule is released. This is a geometric unlocking event, not a simple energy-over-barrier event.^[14] The desorption rate is set by the rate at which this geometric incompatibility condition is met, which depends on the amplitude and frequency of both the adsorbate vibration and the surface phonon modes.

The practical consequence is that desorption rates are sensitive to the specific phonon modes of the surface, not just to the average surface temperature. A surface whose phonon spectrum has modes at frequencies that couple well to the adsorbate's locking vibration will show faster desorption than a surface with the same average temperature but poor phonon-adsorbate coupling. Phonon mode coupling is geometric: it depends on whether the surface phonon displacement pattern has the right symmetry to distort the wake trough in the direction that unlocks the adsorbate. Mode selectivity in thermal desorption is a geometric resonance phenomenon.

5.2 Photodesorption and Non-Thermal Desorption as Wake Resonance

Photodesorption is the most direct evidence that desorption is a wake-field event. In photodesorption, a photon absorbed by the surface — rather than thermal phonon excitation — causes an adsorbed molecule to desorb. The fact that desorption can be triggered by a photon absorbed by the surface, not by the molecule, proves that desorption is mediated by the surface field and not simply by direct energy transfer to the adsorbate.

The mechanism in the DIET (desorption induced by electronic transitions) framework is as follows. A photon absorbed by the surface creates an excited electronic state — typically a hot electron or a hole in the surface electronic structure. This excitation is a localized disturbance in the surface electron density — it is a perturbation of the wake field. The perturbation propagates through the wake field as a wave packet. When that wave packet reaches the site where a molecule is locked into the wake trough, it transiently distorts the

trough geometry. If the distortion is large enough and has the right geometric character, the lock breaks and the molecule desorbs before thermal equilibration can occur.^[15]

The proof that this is a geometric wake event lies in the angular and translational energy distributions of the desorbing molecules. Molecules desorbing via DIET do not have thermal translational energy distributions. Their translational energies are higher than thermal, and they desorb preferentially along specific angular directions relative to the surface normal. Those preferred directions reflect the geometry of the wake field at the desorption site: the molecule is ejected in the direction that the geometric unlocking event imparted momentum, which is determined by the geometry of the trough walls and the direction of the wave packet propagation through the wake field.^[16]

Substrate-mediated photodesorption — where a photon absorbed far from the adsorption site still causes desorption via electron propagation through the wake field — directly demonstrates the extended spatial character of the wake. The hot electron does not stay where the photon was absorbed; it propagates through the surface electronic wake field to wherever the highest wake-gradient sites are, because the gradient-following behavior of the electron wave packet is itself determined by the wake geometry. Molecules locked at high-curvature sites — step edges, kink sites — are the primary targets of substrate-mediated photodesorption because those are the sites where the wake field gradient is steepest and the wave packet is most efficiently channeled.

6. Catalytic Surfaces as Wake-Shaping Boundaries

6.1 The Catalyst's Actual Job

A catalyst lowers the activation energy of a reaction. This statement is true, but it is a thermodynamic description of an outcome, not a mechanism. The mechanism of activation energy lowering by a surface catalyst is geometric: the catalyst's boundary geometry shapes the wake field in a way that locks reactant molecules in a configuration geometrically pre-distorted toward the transition state geometry. The energy cost of reaching the transition

state is reduced because the surface has already done part of the geometric work of deforming the reactant.

Consider the dissociative adsorption of H_2 on a transition metal surface — the first step in most heterogeneous hydrogenation reactions. In the gas phase, the H-H bond must be stretched to its dissociation limit with simultaneous approach of both H atoms to the electron donor geometry of the transition metal. This requires substantial energy: the gas-phase H_2 dissociation energy is 432 kJ/mol. On a Pd(111) surface, H_2 dissociation is nearly barrierless.^[17] The reason is geometric. The Pd(111) surface wake field at the bridge and hollow adsorption sites has a geometry that stretches the H-H internuclear axis as the molecule locks in — the molecule descends into the wake field in a flat-lying configuration and the wake trough geometry at the bridge site pulls the two H atoms apart as the molecule locks deeper. The surface has geometrically pre-distorted H_2 toward the dissociated H + H geometry. The transition state is already substantially achieved by the time the molecule reaches the chemisorption well.

Selectivity in catalysis is wake geometry selectivity. When a surface can lock a molecule in multiple configurations — multiple geometric matches — the product distribution reflects which configuration is most accessible given the wake geometry. A surface whose wake field geometry pre-distorts a prochiral reactant asymmetrically will produce chiral products. Chiral surface catalysis, where a surface modified with a chiral molecule produces enantioselective hydrogenation, works because the chiral modifier reshapes the surface wake field geometry — it introduces an asymmetric geometric bias into the trough that locks the reactant, pre-distorting it toward one enantiomeric transition state over the other.^[18]

Reaction selectivity between competing products — for example, CO oxidation versus CO_2 oxidation, or partial hydrogenation versus full hydrogenation — is determined by the wake geometry at the active site. The product formed is the one whose transition state geometry is best matched by the local wake field. Engineering selectivity without changing composition means changing the wake geometry at the active site — by step density

control, by surface alloying, by strain, or by geometric modification of the support-metal interface.

6.2 Ensemble Effects and Geometric Wake Coupling

Bimetallic catalysts and surface alloys frequently show catalytic activities and selectivities that cannot be predicted from the properties of either pure component. The ensemble effect — the dependence of catalytic activity on the specific arrangement of atom types within the surface layer — is typically interpreted in terms of electronic ligand effects: the presence of one metal type modifies the electronic structure of neighboring metal atoms, changing their binding energies for adsorbates. This interpretation is correct in its consequences but incomplete in its mechanism. The ligand effect is a geometric wake coupling effect.

Two adjacent surface atoms with different nuclear charges have different local potentials. Their individual boundary conditions — the local wake fields they generate — are geometrically distinct. When they sit next to each other in the surface layer, their wake fields overlap and interfere. The interference pattern is not a simple superposition: the wake field oscillations from one atom are phase-shifted relative to those of its neighbor because their Fermi wave vectors are different and their electron spillout geometries are different. The resulting combined wake field at the bimetallic surface site has a geometry that is qualitatively different from either pure component.^[19]

The Pd-Au bimetallic system illustrates this. Pure Au is nearly inert. Pure Pd is active for many hydrogenation reactions but shows poor selectivity for partial hydrogenation. Pd monomers isolated in an Au surface host — where each Pd atom is surrounded entirely by Au atoms — show exceptional selectivity for selective hydrogenation of alkynes to alkenes. The geometric explanation is direct: the Au host atoms' wake field geometry prevents the geometric locking configuration needed for full hydrogenation, because the Au wake field at the bridge sites adjacent to the Pd monomer is geometrically incompatible with the flat-

lying configuration required for further alkene adsorption. The Pd-Au ensemble's combined wake geometry selectively locks alkynes but rejects the alkene product.^[20]

Designing bimetallic catalysts is geometric wake engineering. The practitioner's empirical tool — scanning through bimetallic combinations to find the one with the best selectivity and activity — is implicitly a search through the space of possible wake field geometries produced by different metallic combinations. A geometric framework allows that search to be directed rather than combinatorial: characterize the wake field geometry needed for the target transition state, then identify which bimetallic combination produces a surface wake field that matches it.

6.3 Particle Size Effects as Curvature-Dependent Wake Shaping

Catalytic activity in supported metal nanoparticle systems is a strong function of particle size. For gold, catalytic CO oxidation activity appears sharply below about 5 nm particle diameter and is essentially zero for bulk gold.^[9] For platinum, the oxygen reduction reaction activity per surface atom peaks at intermediate sizes around 3–5 nm and decreases for very small or very large particles.^[21] These size-dependent activity trends are often attributed to quantum confinement effects or to changes in the proportion of low-coordination surface sites. Both are partially correct, and both are expressions of the same underlying change: particle size changes boundary curvature, and boundary curvature determines wake geometry.

A 1 nm metal nanoparticle has a mean surface atom coordination number of approximately 4–5, compared to 9 for a (111) terrace atom on a bulk crystal. Every surface atom on a 1 nm particle is in a high-curvature, low-coordination geometric environment. The wake field of the entire particle is dominated by high-gradient, angular oscillations — there are no large flat terrace regions with uniform Friedel oscillation decay. The wake field is geometrically complex, with high-curvature features everywhere.

A 10 nm nanoparticle has mostly flat (111) and (100) facets, with step edges at facet boundaries. Its wake field is dominated by the flat-terrace Friedel oscillation pattern, with

step-edge perturbations at facet boundaries. This is already close to the bulk surface wake geometry, and its catalytic properties are correspondingly close to bulk properties.

The "magic sizes" observed in nanoparticle catalysis — discrete particle sizes at which activity peaks sharply — are geometric resonances. At a magic size, the particle geometry creates a wake field structure whose spatial frequencies are resonant with the geometric requirements of a specific reaction transition state. Off that size, the resonance is lost. This interpretation is consistent with the observed sensitivity of magic size effects to the specific reaction being studied: a particle size that is resonant for CO oxidation need not be resonant for propylene hydrogenation, because those two reactions have different transition state geometries and therefore different wake field requirements.^[22]

7. Surface Reactivity as Boundary-Curvature Quality

7.1 Curvature as the Master Variable

Every quantitative measure of surface reactivity can be traced to boundary curvature. Binding energy increases with curvature. Reaction barriers decrease with curvature, up to the geometric optimum. Selectivity is a function of local curvature distribution. Turnover frequency correlates with the density of high-curvature sites. These are not independent correlations — they are all expressions of the same underlying mechanism: boundary curvature sets wake field gradient, which sets the depth and shape of the potential trough in which reactant molecules lock, which determines all observable reactivity metrics.

Surface area is not sufficient to predict reactivity, and never has been. A large, flat surface — high surface area, low curvature — can be far less reactive than a small, curved surface with high curvature everywhere. This is why surface-area-normalized turnover frequencies are a more informative metric than simple activity per gram. But even turnover frequency normalized to surface area can miss the point if the surface area measurement does not account for the geometric quality of the surface — its curvature distribution, not just its

total extent. The appropriate metric is curvature-weighted reactive site density, not geometric surface area.

The implication for catalyst design is direct: maximizing surface curvature should be a primary design goal, constrained by thermodynamic stability requirements. The reason this principle is not universally stated in the catalysis literature is that curvature is harder to measure than composition. But the tools exist to measure it — Section 10.1 establishes this — and the engineering strategies to control it exist as well.

7.2 The d-Band Model as Geometric Statement

The d-band center model of Hammer and Nørskov is the most successful quantitative framework for predicting trends in transition metal surface reactivity.^[23] It states that the reactivity of a metal surface toward adsorbates correlates with the position of the d-band center relative to the Fermi level: surfaces with a higher d-band center (closer to the Fermi level) bind adsorbates more strongly and are more reactive. The model successfully explains the trends in hydrogen and CO binding energies across the transition metals, the activity trends in nitrogen dissociation, and the shifts in reactivity induced by alloying and strain.

The standard framing of the d-band model is electronic: the d-band center position determines the anti-bonding orbital occupancy for the adsorbate-surface interaction, which determines the net bond strength. This framing is correct. The geometric restatement is more fundamental: the d-band center position is determined by the geometry of the d-electron density at the surface, which is set by the boundary geometry.

When surface atoms have lower coordination — either due to higher bulk curvature (nanoparticles), surface steps, or the choice of a metal with a narrower d-band — the d-electron density at the surface is more spatially contracted. Contraction of the d-electron density narrows the d-band in energy and pushes its center closer to the Fermi level, because the sum rule on d-band width and d-band filling forces the center upward as the band narrows.^[23] Spatial contraction of electron density is a geometric property: it is the

consequence of the surface boundary condition forcing d-electron wave functions into a more confined geometric space than the bulk interior provides. The d-band model works because d-band center position is a proxy for the geometric quality of the surface boundary — its curvature, its coordination deficit, its wake field gradient strength.

Strain effects on surface reactivity are cleanly understood in this framework. Compressive strain narrows the d-band (atoms are closer together, wave function overlap increases, band broadens — which is the opposite: actually tensile strain narrows the band and raises the center, compressive strain broadens and lowers it). The strain changes the lattice constant, which changes the boundary periodicity, which changes the wake field wavelength and gradient. The d-band center shifts because the geometric boundary condition has changed. The d-band model describes the electronic consequence; the geometric framework describes the cause.

8. Electron Transfer as Wake-Mediated Routing

8.1 The Routing Mechanism

Electron transfer at surfaces — the core event in electrochemistry, photocatalysis, sensor operation, and corrosion — is commonly described as an electron jumping from a donor to an acceptor across an interface. This picture is misleading because it implies a point-to-point process that bypasses the spatial structure of the interface. Electron transfer at surfaces is not a jump. It is a routing event: electron density flows along the high-gradient channels in the wake field that connects the donor site to the acceptor site, following the geometric structure of the wake just as fluid follows pressure gradients.

The surface wake field is a structured medium. Its Friedel oscillations create alternating regions of high and low electron density — in the high-density crests, electron transport is facilitated; in the low-density troughs, it is suppressed. An electron transferring from a surface to an adsorbed molecule does not travel through vacuum — it travels through the spatially structured Friedel oscillation pattern of the wake field. The path it takes and the

rate at which it traverses that path are determined by the geometric quality of the wake channel between the surface donor state and the molecular acceptor state.

This routing picture explains several observations that are awkward in the jump picture. First, electron transfer rates in electrochemistry depend on the crystallographic orientation of the electrode — (111), (100), and (110) faces of the same metal show different exchange current densities for the same redox couple.^[24] The wake field geometry differs for these orientations because the boundary geometry differs. Different wake geometries present different channel structures for electron routing, and therefore different electron transfer rates. The jump picture has no natural explanation for orientation dependence, because if the electron is jumping across a gap, it should not care about what the electrode crystal structure is doing.

Second, the rate of electron transfer is sensitive to the presence of specifically adsorbed anions and solvent molecules at the electrode surface — not just to the average double-layer potential. Specifically adsorbed species modify the surface wake field geometry by locking into the surface at specific geometric sites and thereby distorting the Friedel oscillation pattern in their vicinity. This geometric distortion can either enhance or disrupt the wake channel through which electron transfer occurs. The dependence of electron transfer kinetics on the identity of specifically adsorbed species is a geometric wake disruption effect.

Third, photocatalytic electron transfer — where a photon creates an electron-hole pair in a semiconductor and the electron must transfer to an adsorbed reactant — is strongly dependent on surface faceting. TiO_2 photocatalysis, for example, shows facet-dependent activity where (001) and (101) facets of anatase contribute differently to the overall photocatalytic rate.^[25] The electron and hole generated by photoexcitation route through the crystal to the surface following the geometric channels of the wake field. Different facets present different wake field geometries to the arriving carriers, and those geometries determine whether the carrier reaches the surface efficiently and whether it can transfer to the available adsorbate geometry.

8.2 Electrochemical Interfaces as Geometry-Controlled Wake Junctions

The electrochemical double layer — the interfacial structure at a metal electrode in contact with an electrolyte — is a wake structure. Its two main regions, the compact Helmholtz layer and the diffuse Gouy-Chapman layer, correspond to two distinct regions of the surface wake field.

The compact Helmholtz layer is the region of the wake field where the surface boundary condition dominates all other effects. In this region, the Friedel oscillation amplitude is at its maximum, the electric field is controlled primarily by the surface charge and the specifically adsorbed species geometry, and ion positioning is determined by the geometric match between ion geometry and surface wake trough geometry. Specifically adsorbed anions and water molecules in the Helmholtz layer are geometrically locked into the surface wake in the same way as gas-phase adsorbates — they sit in wake troughs defined by the surface boundary geometry, and their positions and orientations are determined by geometric compatibility between their electron density and the wake field.

The diffuse Gouy-Chapman layer is the outer wake region. Here, Friedel oscillations have damped out and thermal motion competes with geometric ordering. Ion distribution in this layer is well-described by the Poisson-Boltzmann equation — thermal statistics dominate over geometric ordering. The transition from the Helmholtz layer to the diffuse layer is the spatial boundary where the wake field geometry loses its dominance over ion positioning.

The Butler-Volmer equation describes the potential dependence of electrochemical reaction rates through two parameters: the exchange current density (i_0) and the transfer coefficient (α). In the geometric wake framework, i_0 is the rate of electron transfer through the geometric wake channel at zero overpotential — it reflects the geometric quality of the wake junction at the electrode-electrolyte interface. α reflects the geometric asymmetry of the wake potential trough: how the trough shape changes as the electrode potential shifts, and whether the trough deforms more easily in the forward (oxidation) or reverse (reduction) direction.^[26]

Overpotential is the extra potential that must be applied beyond the equilibrium potential to drive an electrochemical reaction at a useful rate. In the wake framework, overpotential is the energy cost of forcing electron transfer through a wake channel whose geometry is mismatched to the reaction's electron density requirements. A geometrically perfect wake channel — one whose spatial structure exactly matches the electron density reorganization required by the reaction — would require zero overpotential beyond the thermodynamic minimum. Real electrodes have imperfect wake channel geometry, and the overpotential quantifies the geometric mismatch. Reducing overpotential means improving wake channel geometry, which means engineering the surface boundary geometry to produce the wake field most compatible with the electron transfer geometry of the target reaction.

9. Failure Modes

9.1 Poisoning as Geometric Blocking

Catalyst poisoning is the loss of catalytic activity due to the adsorption of a species that is not a reactant or product of the intended reaction. The conventional description is site blocking: the poison occupies active sites, reducing the number of sites available for the intended reaction. This description is quantitatively incomplete. Poisons are effective at concentrations far below monolayer coverage — sulfur poisons platinum for SO_2 reduction at coverages of a few percent of a monolayer. If site blocking were the entire mechanism, a 5% poison coverage would reduce activity by at most 5%. In practice, the activity reduction is far larger.^[27]

The geometric wake explanation is this: the poison molecule locks into the surface wake field at a specific site and, by doing so, distorts the wake field geometry in an extended region around that site. Because the wake field is extended — Friedel oscillations propagate over distances of several lattice spacings — a poison molecule at one site modifies the wake geometry at multiple neighboring sites. The neighboring sites, whose wake geometry has been distorted by the poison's locked electron density, no longer present the correct

geometric trough for the intended reactant. Their effective reactivity is suppressed even though they are not occupied by the poison.

CO poisoning of platinum fuel cell catalysts is the clearest example. CO binds strongly to Pt atop sites via deep geometric locking into the surface wake. Each adsorbed CO molecule distorts the Friedel oscillation pattern in its vicinity, changing the wake field geometry at adjacent bridge and hollow sites — the sites where O₂ dissociation and O adsorption proceed in the oxygen reduction reaction. At CO coverages as low as 10–20%, oxygen reduction activity on Pt is suppressed by over 90%.^[27] CO is not occupying 90% of the surface — it is distorting the wake field over 90% of the reactive area. The geometric disruption radius of a single adsorbed CO molecule extends over multiple unit cells.

This geometric understanding of poisoning leads to a design strategy for poison-tolerant catalysts that the site-blocking picture cannot generate: instead of trying to prevent poison adsorption, engineer the surface wake geometry so that the poison's locked geometry does not significantly distort the wake in the vicinity of the intended active sites. If the poison sites and the active sites have geometrically separated wake fields — for example, on a bimetallic surface where the poison preferentially locks on one metal component and the reaction preferentially proceeds on another — the geometric disruption radius of the poison does not reach the active sites. This is precisely why PtRu alloys show improved CO tolerance in fuel cell applications: Ru preferentially adsorbs CO and oxidizes it to CO₂, but the geometric wake disruption from Ru-adsorbed CO is geometrically localized to the Ru sites and does not substantially distort the Pt site wake field.^[28]

9.2 Sintering as Geometric Coarsening

Sintering is the agglomeration of small supported metal nanoparticles into larger ones under thermal conditions. It kills catalytic activity in supported metal catalysts across virtually every industrial application — it is one of the primary deactivation mechanisms in automotive catalytic converters, reforming catalysts, and Fischer-Tropsch synthesis

catalysts. The explanation offered is usually that sintering reduces total surface area, decreasing the number of available active sites. This is correct but incomplete.

Two 1 nm particles that merge into one 2 nm particle do lose total surface area — approximately by a factor of two in total, since surface area scales as r^2 while volume scales as r^3 . But the activity loss is typically larger than the surface area loss. The geometric wake explanation identifies why: sintering reduces curvature, not just area. The two 1 nm particles each had high-curvature surface geometry everywhere — every surface atom was a high-curvature site. The merged 2 nm particle has lower average curvature, because the high-curvature tip regions of each original particle are now buried in the interior of the merged particle. The surface that remains is dominated by lower-curvature facet geometry.

Lower curvature means weaker wake field gradients, shallower potential troughs, weaker geometric locking of reactants, and higher reaction barriers. Activity per remaining surface site decreases as curvature decreases, not just activity per gram. Temperature-stable catalysts must therefore be geometrically stabilized against curvature loss, not simply chemically stabilized against bulk phase change. Strategies include strong metal-support interactions that pin particle geometry by anchoring high-curvature features to the support interface, core-shell structures that geometrically confine the active metal surface, and encapsulation architectures that physically prevent particle contact and merger.^[29]

9.3 Surface Reconstruction under Reaction Conditions as Geometric Drift

A surface characterized under ultra-high vacuum before a catalytic experiment does not necessarily retain that geometry under reaction conditions. Adsorbates locked into the surface wake field exert back-pressure on the surface geometry — they deform the surface lattice through the same wake-field coupling that determines their binding. At high adsorbate coverages or with strongly binding adsorbates, this back-pressure can be sufficient to drive surface reconstruction. The surface geometry drifts from its clean-surface optimum to a new geometric configuration stabilized by the adsorbate-surface geometric coupling.

This is not a hypothetical concern. CO adsorption on Pt(100) drives a reconstruction from the (1×1) bulk-truncated geometry to a hex-reconstructed surface at elevated CO pressures. N adsorption on Fe surfaces used in ammonia synthesis drives reconstruction of the Fe surface toward geometries with higher density of C7 sites — the sites most geometrically active for N₂ dissociation.^[30] Oxide surfaces under oxidizing or reducing conditions can undergo major geometric restructuring that completely changes the wake field and therefore the catalytic behavior.

This geometric drift is why operando characterization is essential, not optional. Ambient-pressure X-ray photoelectron spectroscopy (AP-XPS), in-situ transmission electron microscopy (TEM), and surface X-ray diffraction under reaction conditions capture the actual boundary geometry during catalysis, rather than the pre-reaction geometry that traditional ex-situ methods measure. The pre-reaction geometry may bear no quantitative relationship to the working geometry, and designing catalysts based on the pre-reaction geometry alone is designing for a condition that does not exist during operation.

10. Restoring Boundary and Wake Geometry to Fix Surface Chemistry

10.1 Diagnosis First: Measure the Actual Geometry

If surface chemistry is boundary geometry, then diagnosing a surface chemistry problem requires measuring the actual boundary geometry — not the nominal composition, not the average crystal structure, not the BET surface area. The following techniques each provide distinct geometric information about the surface boundary condition and its wake field expression.

Low-energy electron diffraction (LEED) measures the surface unit cell periodicity and, through LEED-I(V) analysis (the dependence of diffraction spot intensities on electron beam energy), the interlayer spacings and atomic positions in the top several layers. LEED-I(V) analysis directly reveals surface relaxation and reconstruction — the geometric

adjustment of the boundary from the bulk-truncated structure. It provides sub-angstrom resolution of the average geometric structure over the illuminated area, typically a region of order 1 mm^2 . Its limitation is that it averages over defects, steps, and local geometric variations.^[31]

Scanning tunneling microscopy (STM) and atomic force microscopy (AFM) provide real-space images of the surface geometry with atomic resolution. STM images the local density of states at the surface — which is a direct map of the wake field at the Fermi level — and therefore provides geometric information about both the atomic positions and the electronic wake structure simultaneously. AFM provides geometric force maps that are sensitive to the local curvature of the surface potential. Together, STM and AFM can map the curvature distribution of a real surface — identifying step densities, kink densities, adatom concentrations, and domain boundaries.

Angle-resolved photoemission spectroscopy (ARPES) maps the electronic band structure of the surface, including the surface electronic states that are the direct expression of the wake field's electronic component. ARPES provides the k-space structure of the surface wake — its Fourier decomposition in the parallel momentum directions. From ARPES data, the Fermi wave vector k_F and the d-band center position are directly measurable. These quantities set the Friedel oscillation wavelength and the wake field gradient strength, respectively. ARPES on stepped surfaces resolves the modulation of surface states by step periodicity — it directly images the lateral structure of the wake imposed by the step geometry.^[32]

Synchrotron-based surface X-ray diffraction (SXRD) provides sub-angstrom precision in surface atomic positions for both ordered and partially disordered surfaces, and it can be performed in gas or liquid environments — making it the premier technique for operando geometric characterization. Crystal truncation rod measurements in SXRD directly access the interlayer spacings and in-plane atomic positions of the surface layers, yielding the complete boundary geometry description. Combined with ab initio calculations, SXRD

data can be used to compute the expected Friedel oscillation structure and wake field gradient at the measured surface geometry.

10.2 Geometric Engineering Strategies

Once the surface geometry is diagnosed, it can be engineered. The following strategies address boundary geometry directly, each targeting a different geometric variable.

Step engineering controls the density and geometry of step edges on single-crystal and polycrystalline surfaces. By cutting a single-crystal surface at a controlled miscut angle from a low-index plane, a regular array of monoatomic steps is produced, with terrace width inversely proportional to the miscut angle. The step density — and therefore the density of high-curvature, high-gradient wake sites — is tunable continuously by adjusting the miscut angle. Experiments on nitrogen dissociation on Ru surfaces have shown that step density directly controls the dissociation rate, with stepped Ru(109) being orders of magnitude more active than flat Ru(0001) for N₂ dissociation under comparable conditions.^[4] Step engineering translates directly into wake gradient engineering.

Nanoparticle shape control through colloidal synthesis stabilizes specific crystallographic facets — and therefore specific wake geometries — on the surfaces of catalytic nanoparticles. Capping agents in colloidal synthesis bind selectively to specific surface facets, lowering their surface energy and stabilizing them relative to other facets during growth. By choosing capping agents that bind to the facets with the desired wake geometry, the synthesis produces particles whose surfaces are predominantly that facet. Pt nanocubes with predominantly (100) facets have been synthesized and show different activity than Pt nanospheres with mixed faceting — the geometric difference in wake field between (100) and (111) facets produces the activity difference.^[33]

Surface alloying and intermetallic compound formation introduce a second component into the surface layer at controlled concentrations and arrangements. Unlike bulk alloying, which distributes both components throughout the material, surface alloying concentrates the geometric wake coupling effect at the surface where it is needed. Vapor deposition of

small amounts of a second metal followed by controlled annealing can produce surface alloys with specific geometric arrangements — random solid solutions, ordered surface intermetallics, or Janus-type segregated structures — each with a distinct wake field geometry. The geometric target — the wake field needed for the desired reaction — should drive the choice of alloying partner and arrangement.

Strain engineering modifies the surface lattice constant by growing thin films on substrates with controlled lattice mismatch, or by mechanical deformation of thin-film electrodes. Epitaxial compressive or tensile strain changes the boundary periodicity of the surface lattice, which changes the Friedel oscillation wavelength and the d-band center position. Pt monolayers on Pd substrates are under compressive strain, which downshifts the Pt d-band center and weakens oxygen binding — beneficial for the oxygen reduction reaction where over-binding of oxygen intermediates is a limiting factor on Pt.^[34] Strain engineering is wake-wavelength engineering: it tunes the spatial frequency of the surface wake field to better match the geometric requirements of the target reaction.

Passivation and selective depassivation use adsorbate species to block or restore specific wake channels. Thiols on gold surfaces block virtually all surface reactivity by locking into the gold surface wake at all available atop sites and creating a dense, well-ordered overlayer that presents a geometrically inert, alkyl-terminated outer surface. Selective depassivation — using electrochemical oxidation, localized plasma treatment, or photochemical removal — can reopen specific geometric sites while leaving others passivated, creating surfaces with spatially patterned wake geometries. This is the geometric basis for self-assembled monolayer-based surface chemistry.

10.3 The Design Principle

State the design principle without qualification: design the boundary geometry to produce the wake field that locks reactants in the configuration geometrically closest to the transition state, routes electrons through the lowest-barrier wake channel, and releases products cleanly without deep re-locking. This is the complete design specification for a

surface-active material. Everything else — composition selection, doping strategy, support choice, synthesis temperature and atmosphere, post-synthesis treatment — is in service of achieving and maintaining that geometric configuration.

When a surface chemistry system fails — activity decays, selectivity drifts, overpotential increases, sensitivity drops — the first diagnostic question is not "what is the composition?" It is "what did the geometry do?" Did the active sites sinter and lose curvature? Did the surface reconstruct under reaction conditions to a less reactive geometry? Did a poison disrupt the wake field geometry across a large area? Did strain relax and remove the tuned lattice constant that set the wake wavelength? Once the geometric failure mode is identified, the fix is geometric. The composition question comes second, only to the extent that composition is the tool used to achieve the required geometry.

11. Discussion

The geometric wake framework presented in this paper is not a replacement for the quantitative models that currently dominate surface chemistry and heterogeneous catalysis. The d-band center model, the Brønsted-Evans-Polanyi (BEP) relations, the Sabatier volcano, and Marcus theory for electrochemical electron transfer are all quantitatively successful within their domains of applicability. The geometric framework provides the physical basis that explains why they are successful and identifies where and why they fail.

The d-band center model works because d-band center position is a proxy for surface boundary geometry. As established in Section 7.2, the d-band center shifts because the geometry of the d-electron density at the surface shifts — contracted d-electrons (high curvature, low coordination) produce a high d-band center; expanded d-electrons (low curvature, high coordination) produce a low d-band center. The model breaks down when two surfaces have the same d-band center position achieved by different geometric routes — for example, a surface under compressive strain versus a surface with increased step

density can both show an upshifted d-band center, but their reactivity patterns can differ because the geometric structure of their wake fields differs even though the integrated wake strength (d-band center) is the same. The d-band model is insensitive to the angular and spatial structure of the wake — it collapses that structure to a single scalar number — and it fails when that angular structure is the determining factor for a specific reaction. ^[23]

The Brønsted-Evans-Polanyi relations express a linear correlation between reaction energy and activation energy across a set of related surface reactions. They work because, for a family of structurally similar transition states, the transition state geometry is a fixed fractional deformation of the reactant geometry — the geometric path from reactant to transition state is approximately the same for all members of the family. A surface that pre-distorts the reactant geometry more (stronger geometric locking) has a lower activation energy and also a larger reaction energy — hence the linear correlation. The BEP relation breaks down when the transition state geometry changes qualitatively between members of the family, or when the geometric locking mechanism changes — as happens for reactions on structurally different surface sites (terrace vs. step) where the wake geometry changes the geometric path to the transition state. ^[35]

The Sabatier volcano and its modern computational implementation — the activity volcano plots of Nørskov and colleagues — identify the optimal binding energy for a given reaction. The volcano works because binding energy is a scalar measure of wake field depth, and the optimal wake depth (the volcano peak) is the one that best geometric-matches the reactant through the transition state to the product. The volcano breaks down when a single binding energy descriptor is insufficient to characterize the multi-dimensional geometric compatibility between the wake field and the full reaction pathway — which occurs for reactions with multiple intermediates whose geometric requirements are not all correlated with the binding energy of a single descriptor molecule. ^[13]

Marcus theory for electrochemical electron transfer introduces the reorganization energy λ as the key parameter controlling electron transfer kinetics. In the geometric wake framework, λ is the energy cost of the geometric reorganization that the wake field must

undergo to accommodate the transition from reactant to product electron density geometry. Surfaces with better geometric wake-channel quality require less reorganization to route electron density from donor to acceptor, and therefore show smaller λ and faster electron transfer rates. Marcus theory breaks down in the inverted region for surface-mediated reactions more readily than for homogeneous reactions, because the surface wake field can couple the electron transfer to surface phonon modes in ways that have no homogeneous analog — a geometric coupling that is absent from the original Marcus formulation.^[26]

What the geometric framework does not yet deliver is quantitative prediction of wake geometry from first principles for complex, multi-component surfaces under realistic reaction conditions. Density functional theory (DFT) can compute the electronic structure of a surface with a specified geometry, and from that electronic structure the Friedel oscillation pattern can in principle be computed. But DFT calculations for surfaces under reaction conditions — with multiple adsorbates, at finite temperature, on surfaces that are reconstructing in real time — remain computationally expensive and do not yet provide the full time-resolved wake dynamics that characterize a surface during a catalytic cycle. The connection between mesoscale geometric features — particle size distributions, support morphology, inter-particle wake coupling — and atomic-scale wake structure also requires more experimental validation. Operando methods have advanced rapidly, but time-resolved wake dynamics during a single catalytic turnover remain largely inaccessible with current techniques. These are open problems, and they define the frontier of the geometric wake approach to surface chemistry.

12. Conclusion

Surface chemistry is boundary geometry. This is not a metaphor and it is not an approximation — it is the physical mechanism that underlies every phenomenon the field studies. A surface is a geometric boundary condition imposed on the electron density and

vibrational field of the material. That boundary condition generates a wake field — structured oscillations in electron density and in the phonon field — whose spatial geometry is a direct map of the boundary geometry. That wake field is the medium through which the surface communicates its identity to everything it interacts with.

Adsorption is geometric locking of an incoming molecule's electron density into the wake field trough. Desorption is geometric unlocking driven by thermal phonon or photon excitation of the wake. Catalysis is geometric pre-distortion of reactant geometry toward transition state geometry, achieved by engineering the boundary geometry to shape the wake field appropriately. Electron transfer is routing of electron density through high-gradient wake channels to the acceptor site. Poisoning is geometric disruption of the wake field over an extended area by a single locked contaminant. Sintering is geometric coarsening — loss of boundary curvature and therefore loss of wake gradient and reactivity. Surface reconstruction under reaction conditions is geometric drift — the boundary geometry shifting away from its characterized value in response to adsorbate back-pressure on the wake field.

The existing quantitative models that successfully predict surface reactivity trends — d-band model, BEP relations, Sabatier volcano, Marcus theory — all work because they are proxies for boundary wake geometry. The geometric framework explains why they work and predicts where they fail. The diagnostic tools to measure boundary geometry with sufficient resolution are available now. The synthesis and engineering strategies to control boundary geometry — step engineering, shape-controlled nanoparticle synthesis, surface alloying, strain engineering, passivation — are available now.

The barrier to applying this framework is conceptual. The field has been organized around describing outcomes — binding energies, rate constants, selectivities — rather than tracing those outcomes to the geometry that produces them. Drop the outcome-first language. Follow the geometry. Every broken surface chemistry system has a geometric failure mode. Fix the geometry, fix the chemistry.

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